

Questions		Answers	Marks
9	(a)	Using $Q = mc\Delta\theta$, we get $1.79 \times 10^4 = m \times 5190 \times (50.0 - 10.0)$, giving $m = 0.08622 \text{ kg}$	M1
		number of moles = $\frac{86.22 \text{ g}}{4 \text{ g mol}^{-1}} = 21.56 \text{ mol}$	M1
		number of atoms = $21.56 \times 6.02 \times 10^{23} = 1.298 \times 10^{25} = 1.30 \times 10^{25}$	A1
	(b)	(i) gas that obeys $pV = nRT$	B1
		p : pressure of the gas V : volume of the gas n : number of <u>moles</u> of the gas T : <u>thermodynamic</u> temperature of the gas or temperature in Kelvin	B1
		(ii) for ideal gas, internal energy consists <u>only of kinetic energies</u> of gas molecules.	B1
		$\Delta u = n \left[\frac{3}{2} R (T_f - T_i) \right] = 21.56 \times \frac{3}{2} \times 8.31 \times -40 = -10749 = -1.07 \times 10^4 \text{ J}$ allow $\Delta u = \frac{3}{2} n (p_f V_f - p_i V_i)$ if the volumes of the gas were found. If candidates obtain $\Delta u = 1.07 \times 10^4 \text{ J}$, award maximum of 1 mark even though first B1 may not be awarded.	B1
		(iii) Since $\langle E_k \rangle = \frac{1}{2} m v_{rms}^2 = \frac{3}{2} kT$, $v_{rms} \propto \sqrt{T}$	B1
		$\frac{\sqrt{10 + 273.15} - \sqrt{50 + 273.15}}{\sqrt{50 + 273.15}} \times 100\% = -6.39\%$	B1

	(c)	(i)	<u>increase</u> in internal energy of the system is equals to the sum of thermal energy <u>supplied into</u> the system and work done <u>on</u> the system. B1: for directions of change (underlined)	B2
		(ii)	$\Delta u = q + w$ gives $w = -1.07 \times 10^4 - (-1.79 \times 10^4) = 0.72 \times 10^4 \text{ J}$	B1
			so work done by the gas is $-7.2 \times 10^3 \text{ J}$	B1
		(iii)	for changes at constant volume, $w = 0$, so	B1
			$q = \Delta u = -1.07 \times 10^4 \text{ J}$	B1
	(d)	(i)	Using $V = \frac{nRT}{p}$ gives $V_i = \frac{21.56 \times 8.31 \times (50 + 273.15)}{300\,000} = 0.193 \text{ m}^3$ $V_f = \frac{21.56 \times 8.31 \times (10 + 273.15)}{300\,000} = 0.169 \text{ m}^3$	B1
			pressure / 10^5 Pa 3.00 0.169 0.193 volume / m^3	B1
		(ii)	(positive) work is done on gas when process occurs at constant pressure, however - no work done on gas when volume is constant	B1
			So for Δu to be equal <u>more</u> thermal energy must be released by gas when the process occurs at constant pressure than at constant volume	B1

10	(a)	(i)	A wave in which energy is carried from one point to another by	B1
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